

(1471 Words)

Navigating Ethical Challenges posed by Artificial Intelligence

“All changed, changed utterly: A terrible beauty is born.” This statement by poet William Yeats is an apt way to describe technology today, particularly Artificial Intelligence, which, like fire, is neither good nor bad; its value depends on how we use it.

Artificial intelligence (AI), once the subject of people’s imaginations and the fancy fantasy gadget used in StarTrek, Star Wars, or James Bond movies, is no longer a fantasy of scriptwriters' imagination but rather a commonplace reality of people’s daily lives. What is AI? One of the simplest and generic explanations states that it is “the ability of a computer to mimic the competencies of the human mind.” (Thormundsson, Topic: Artificial Intelligence). AI focuses on creating systems or machines to automate tasks typically performed by humans. This idea of autonomous workers has been thought of “[e]ver since Homer wrote of mechanical ‘tripods’ waiting on the gods at dinner, imagin[ing] mechanical assistants [to be] a part of our culture.” (Buchanan) AI can be attributed to a Roomba but also to an autonomous warhead. Advanced AI, like an autonomous warhead, is part of a subtopic of AI called Machine Learning (ML). AI, while simulating a human mind, can continually improve, using data generated during each subsequent use or iteration past the initial training phase. This is akin to the saying used for humans that with age and experience comes knowledge and wisdom. Deep Learning (DL) is a type of Artificial Intelligence that mimics how human brains work and “generates its knowledge... without injunction of additional data” (Das, Paragraph 10). DL machines are the most advanced and independent AI systems and are “now widely applied in many fields and research areas” (Sarker, Paragraph 4).

While we tend to view AI on a shining pedestal, the reality is, it is far from perfect. Despite AI's many benefits, a prevailing issue is with the ethical use of this technology. To reap the full benefits of Artificial Intelligence, we must address the concerns of biased training data, AI's limited perspective on ethical decision-making, and the lack of human control.

The first ethical dilemma of AI is biased training data. ProPublica, a company specialized in using AI in the justice system, created an algorithm that would give “a rating of [criminals] likelihood of re-offending.” (Abelson, 275). ProPublica programmers trained the algorithm on past cases of convicted criminals who had or had not committed additional crimes. To generalize repeat offenders, the programmers gave the algorithm information about the criminals, like their weight, height, highest education, and race. The algorithm was deployed to police stations across the nation. In 2014, Brisha Borden was running late and rode an unlocked scooter she found on the street to save some time. At the same time, Vernon Prater was caught shoplifting at Home Depot. Borden and Prater were convicted of their crimes, and their cases were fed into the AI system during the booking process. “Borden, a black woman who had one juvenile misdemeanor charge, was deemed high risk, and Prater, a white man with an armed robbery conviction and another pending charge, was deemed low risk” (Abelson, 275). Using biases fed to it by previous cases, the AI system produced results based on stereotypes far from the truth. “Prater, the white man fingered by the computer as low risk, was re-arrested for a major electronics theft and is serving eight years in prison. Borden, identified as high risk, has avoided further legal trouble” (Abelson, 276). Not only are pre-existing stereotypes magnified by AI systems, but training data can easily be manipulated to create stereotypes. With the growing concerns about the biases rooted in AI and the unwillingness of companies to address them, these projected biases and stereotypes are the first ethical concerns of AI systems.

The second ethical dilemma is AI's limited perspective of ethical decision-making. Picture yourself walking down a sidewalk. As you stroll on, you see an autonomous car driving down a road. Suddenly, the car in front of the autonomous car brakes, leaving the autonomous car with a short amount of time to decide how best to minimize casualties. In this scenario, the autonomous car has two options: swerve into oncoming traffic, possibly hitting a car with one passenger, or colliding with the car in front with three passengers. This scenario is best described as an ethical dilemma. Similar to the famous "trolley problem," there is no definitive "correct" answer. Is it more morally correct to hit the car in front of you or the car coming in the oncoming traffic lane? Similarly, is it better to run over a grandma instead of a baby? AI's limited perspective on ethical decision-making is the second ethical dilemma of Artificial Intelligence.

The third ethical dilemma is the lack of transparency behind AI's decisions. A black box is "an electronic recording device for the purpose of facilitating the investigation of accidents and incidents" (Flight). A black box system receives "inputs and produces outputs but its workings are unknowable" (Kenton). While black boxes tend to be inside aircrafts, this terminology is used to describe AI systems "[w]hen focusing on transparency in the context of AI" (Larsson). AI systems are known as black box systems, similar to black boxes in aircrafts because they take inputs without an outsider knowing the internal interactions of how that AI system generates a certain output. Lack of transparency is a significant concern of advanced AI systems because programmers cannot extrapolate data points of how a system reached a certain conclusion. "Transparency and explainability are identified as key quality requirements of AI systems and are portrayed as quality requirements that need more focus in the machine learning context" (Balasubramaniam).

Artificial Intelligence is not just a glimpse into the future; it is the beating heart of our present, shaping how we live, work, and innovate today. “[AI] has allowed for remarkable improvements in numerous fields. For instance, AI applications are analyzing images to detect potentially cancerous cells, helping to predict where and when the next big earthquake [might] strike and allowing for the development of companion robots supporting overburdened caretaking staff” (Buiten). However, the future of AI will depend on effective governance and oversight. The best way to address these ethical issues is through regulation, which will ensure a safe environment for users.

The U.S. Association for Computing Machinery (ACM), the world’s largest educational and scientific computing society, proposed thirteen principles that programmers must consider to tackle the ethical challenges AI poses. Of these thirteen principles, six are deemed “core” principles, and all programmers should follow them when building AI systems.

The first of these principles is personal data control: “Generative AI systems should allow a person to opt out of their data being used to train the system or facilitate its generation of information.” (ACM, Paragraph 9). This principle allows users to choose whether they want their data collected to train the AI system. The second principle is transparency: “In particular, where generative AI is being used to simulate human agents, at all times individuals must be promptly and clearly informed that they are interacting with a system as opposed to a human.” (ACM, Paragraph 11). This principle ensures that users of applications know whether AI systems are being used and that these AI systems are not always correct. The third principle is auditability and contestability: “Providers of Generative AI systems should ensure that system models, algorithms, data, and outputs can be recorded where possible, so that they may be audited and/or contested in appropriate cases” (ACM, Paragraph 12). This principle ensures that AI’s decisions

are accessibly documented to be examined and changed. The fourth principle is interpretability and explainability: “Managers of algorithmic systems are encouraged to produce information regarding both the procedures that the employed algorithms follow (interpretability) and the specific decisions that they make (explainability)” (ACM, Paragraph 18). This principle guarantees that users know how AI systems gather their data and the possible biases in said systems. The fifth principle is maintainability: “Evidence of all algorithmic systems’ soundness should be collected throughout their life cycles” (ACM, Paragraph 19). This principle ensures the upkeep of AI systems. The sixth principle is accountability and responsibility: “Public and private bodies should be held accountable for decisions made by algorithms they use.” (ACM, Paragraph 20). This principle holds both algorithms and programmers responsible for the decisions they make.

The principles proposed by the ACM provide a good framework for programmers, corporations, and government entities to ensure their AI systems are ethical and transparent. By following these principles, AI technology deployments can be safe, fair, constructive, and beneficial. If the Foreign Service develops and enforces sound regulations to address ethical challenges that arise, we can create a future where Artificial Intelligence is not just intelligent but also responsible and trustworthy.

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